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# Certified Lean Six Sigma White Belt (CLSSWB) Training

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WSQ Course Code: TGS-2025053210

Conducted by Tertiary Infotech Academy Pte Ltd · UEN: 201200696W

TSC: Quality Process Control · ELE-QUA-5006-1.1

Version v5 · 23 September 2026 · Trainer: Dr. Alfred Ang

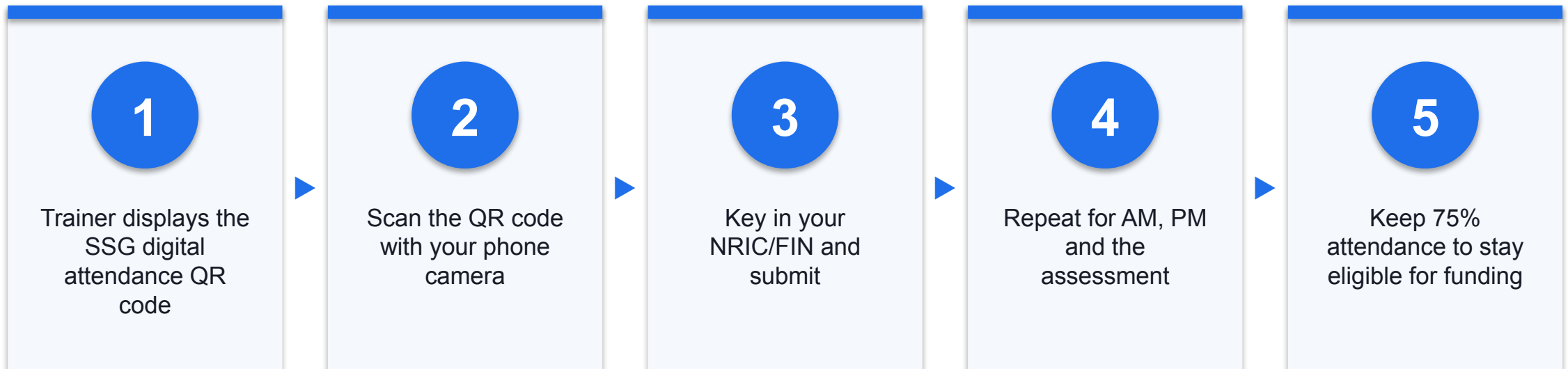
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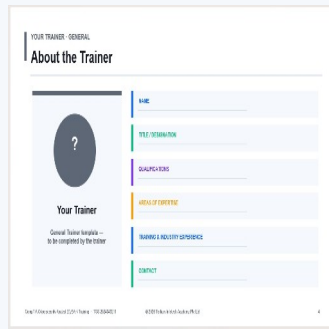
COURSE ADMINISTRATION

# Welcome & Housekeeping

# Digital Attendance (Mandatory)



# About the Trainer



## Your Trainer

General Trainer template —  
to be completed by the trainer

NAME

TITLE / DESIGNATION

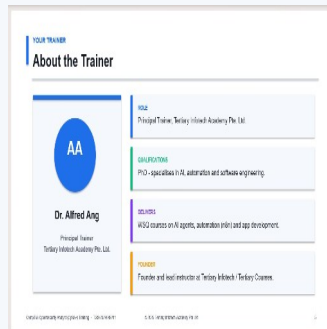
QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

# About the Trainer



## Dr. Alfred Ang

Principal Trainer  
Tertiary Infotech Academy Pte. Ltd.

### ROLE

Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

### QUALIFICATIONS

PhD; Certified Lean Six Sigma practitioner and trainer.

### DELIVERS

WSQ courses on Lean Six Sigma, quality management and data analytics.

### EXPERIENCE

Process improvement across manufacturing, service and technology sectors.

### FOUNDER

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

# Let's Know Each Other

- Your name, organisation and role.
- Your experience with process improvement or quality work (if any).
- One process at work that frustrates you — we may use it as your course scenario.

# Ground Rules

1

Set your mobile phone to silent mode.

2

Participate actively — no question is too small.

3

Mutual respect: agree to disagree.

4

One conversation at a time.

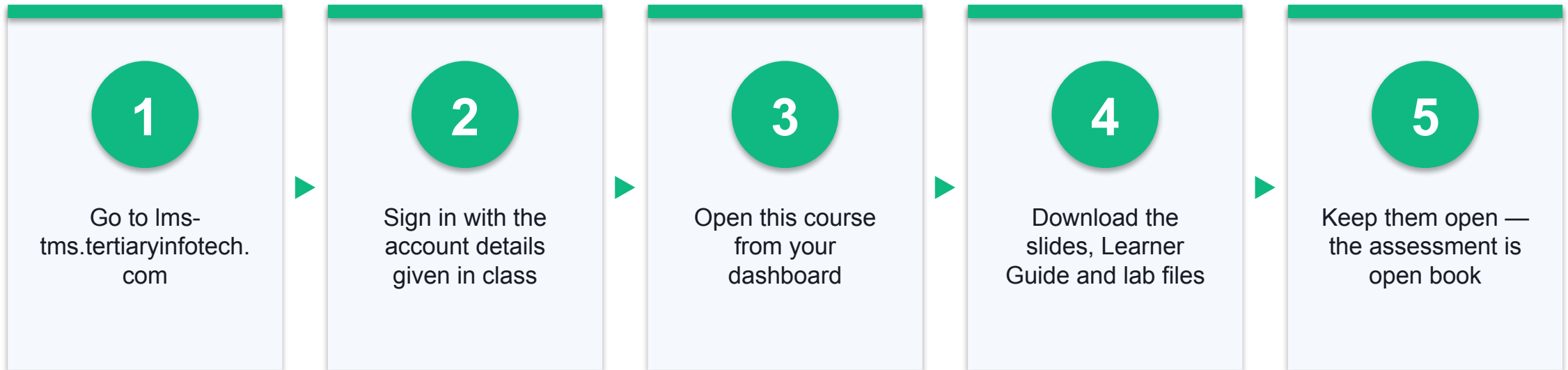
5

Be punctual; return from breaks on time.

6

75% attendance is required for certification.

# Download Your Course Material





# Lesson Plan — 1 Day, 8 Hours

## Morning

- Morning — Foundations and Define
  - Digital attendance (AM) · Introductions
  - Foundations: Quality, Lean, Six Sigma, Lean Six Sigma
  - Belt roles and the DMAIC roadmap
  - DEFINE: VOC, CTQ, problem statement, SMART goal
  - Activity 1 — Define: requirements and problem statement
  - MEASURE: process mapping, SIPOC, data, the 8 wastes
  - Activity 2 — Measure: map the process, spot the waste

## Afternoon

- Afternoon — Analyze, Improve, Control
  - Digital attendance (PM)
  - ANALYZE: root cause, 5 Whys, Fishbone, Pareto
  - Activity 3 — Analyze: find the root cause
  - IMPROVE: solutions, 5S, mistake proofing, piloting
  - Activity 4 — Improve: choose and pilot a countermeasure
  - CONTROL: control plan, visual management, handover
  - Activity 5 — Control: hold the gain · Revision
  - Briefing · Final Assessment (WA + CS)

# Skills Framework — TSC: Quality Process Control

- TSC Code: ELE-QUA-5006-1.1
- A1: Support the definition of an improvement project to meet process performance.
- A2: Assist in establishing the scope of work based on organisational requirements.
- A3: Assist in analysing process performance data to identify causes of variation.
- A4: Support the measurement of process performance against defined quality standards.
- A5: Support improvement and control actions that sustain process performance.
- K1: Basic Lean and Six Sigma concepts — quality, value, waste and variation.
- K2: Basic quality tools used at each phase of the DMAIC roadmap.

# Learning Outcomes

1

## LO1 — Core concepts

Describe quality, Lean, Six Sigma and the belt roles in an improvement team.

2

## LO2 — The DMAIC roadmap

Explain the five phases of DMAIC and what happens in each.

3

## LO3 — Define

Identify customer requirements and a clear problem statement.

4

## LO4 — Measure

Describe how a process is mapped and measured, and identify the 8 wastes.

5

## LO5 — Analyze

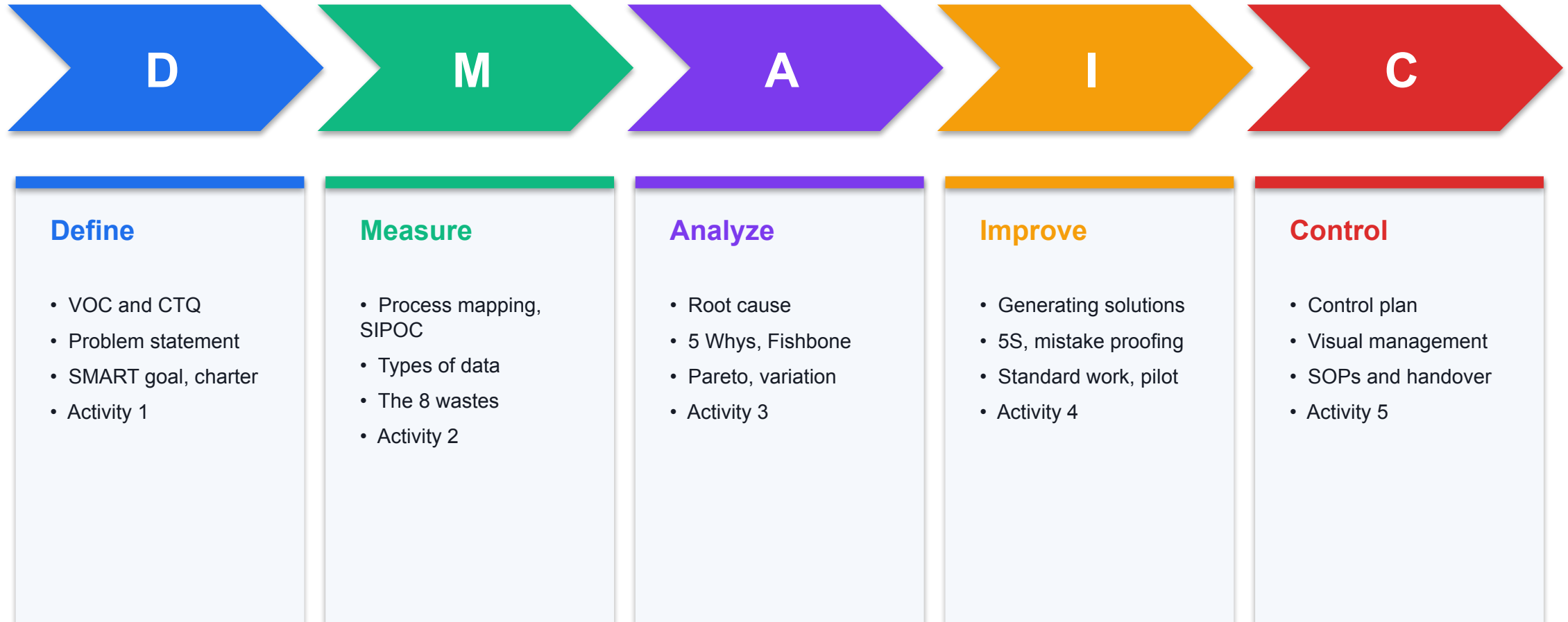
Identify likely causes using 5 Whys and Fishbone analysis.

6

## LO6 — Improve & Control

Describe actions that fix a cause and hold the gain.

# Course Outline — We Follow DMAIC End to End



# Briefing for Assessment

1

## **Phones away**

Place phones and other materials under the table or on the floor.

2

## **No recording**

No photos or recording of assessment scripts.

3

## **Work alone**

No discussion of any kind during the assessment.

4

## **Black or blue pen**

Use a black or blue pen for hard-copy assessments.

5

## **No correction fluid**

No liquid paper or correction tape on your script.

6

## **Pens down on time**

Scripts are collected as soon as time is up.

# Assessment

1

## Written Assessment (WA)

Written Assessment (WA) — Short-Answer Questions (SAQ), 30 minutes, open book.

2

## Case Study (CS)

Case Study (CS) — applied Lean Six Sigma scenario tasks, 30 minutes, open book.

3

## Open book

Slides, Learner Guide, your lab outputs and approved materials only.

4

## Attendance

A minimum of 75% attendance is required to be eligible for assessment and funding.

5

## Result

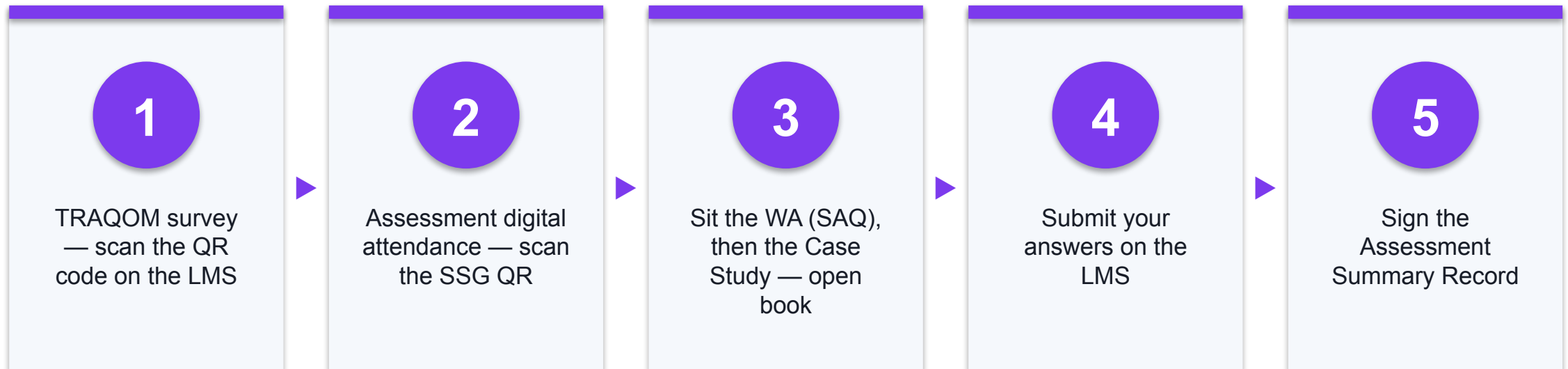
Graded Competent (C) or Not Yet Competent (NYC) by the assessor.

6

## Appeals

An appeal process is available if you wish to contest a result.

# Assessment Flow



FOUNDATIONS

# Six Sigma Foundations

Quality · Lean · Six Sigma · Lean Six Sigma · Belt roles · The DMAIC roadmap



# What is Quality?

Before we can improve quality, we have to agree what the word actually means.

# Three Common Answers — and Why Only One Holds Up

## Defect-free output

"Quality means zero defects"

- Necessary, but not sufficient.
- A product can be flawless and still not sell.
- Defect-free against WHOSE standard?

## Meeting set standards

"Quality means meeting the spec"

- Better — it is measurable.
- But a spec can be set at the wrong level.
- Meeting a spec nobody wants is waste.

## Meeting customer expectations

"Quality means the customer is satisfied"

- This is the Six Sigma definition.
- The customer sets the standard.
- Everything else follows from this.

## FOUNDATIONS · LEAN

# What is Lean?

Lean maximises customer value by systematically removing everything the customer would not pay for.

## WHAT IS LEAN?

# Lean — The Core Idea

1

### Origin

The Toyota Production System — refined in Japanese manufacturing from the 1950s.

2

### Focus

Speed and flow: shorten the time from customer request to delivery.

3

### Enemy

Waste (muda) — any effort that consumes resources but creates no customer value.

4

### Method

See the process, find the waste, remove it, then standardise the improvement.

5

### Applies to

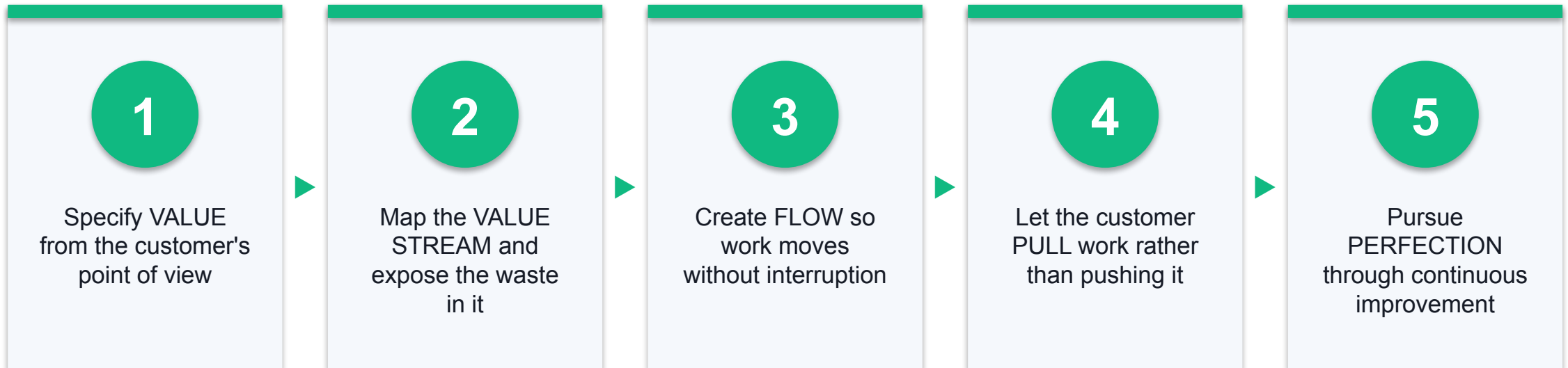
Manufacturing AND service — hospitals, banks, IT service desks, government.

6

### White Belt use

Recognising waste, understanding process maps and supporting 5S in your own area.

# The Five Lean Principles



## FOUNDATIONS · SIX SIGMA

# What is Six Sigma?

A disciplined, data-driven method to reduce variation and defects — so the process delivers the same result every time.

# Six Sigma — The Core Idea

1

## Origin

Motorola, 1986 — later made famous by General Electric under Jack Welch.

2

## Focus

Consistency: reduce the variation that creates defects.

3

## Enemy

Variation — the spread that turns a good average into an unreliable experience.

4

## Method

DMAIC, driven by data and statistics rather than opinion.

5

## The target

No more than 3.4 defects per million opportunities (DPMO).

6

## White Belt use

Helping collect data and taking part in root cause discussions.

# Lean vs Six Sigma vs Lean Six Sigma

## Lean

- Removes waste
- Improves speed and flow
- Question: is this step worth doing?
- Tools: VSM, 5S, Kaizen, poka-yoke



## Six Sigma

- Reduces variation
- Improves consistency and accuracy
- Question: why does the result vary?
- Tools: DMAIC, Pareto, SPC, root cause

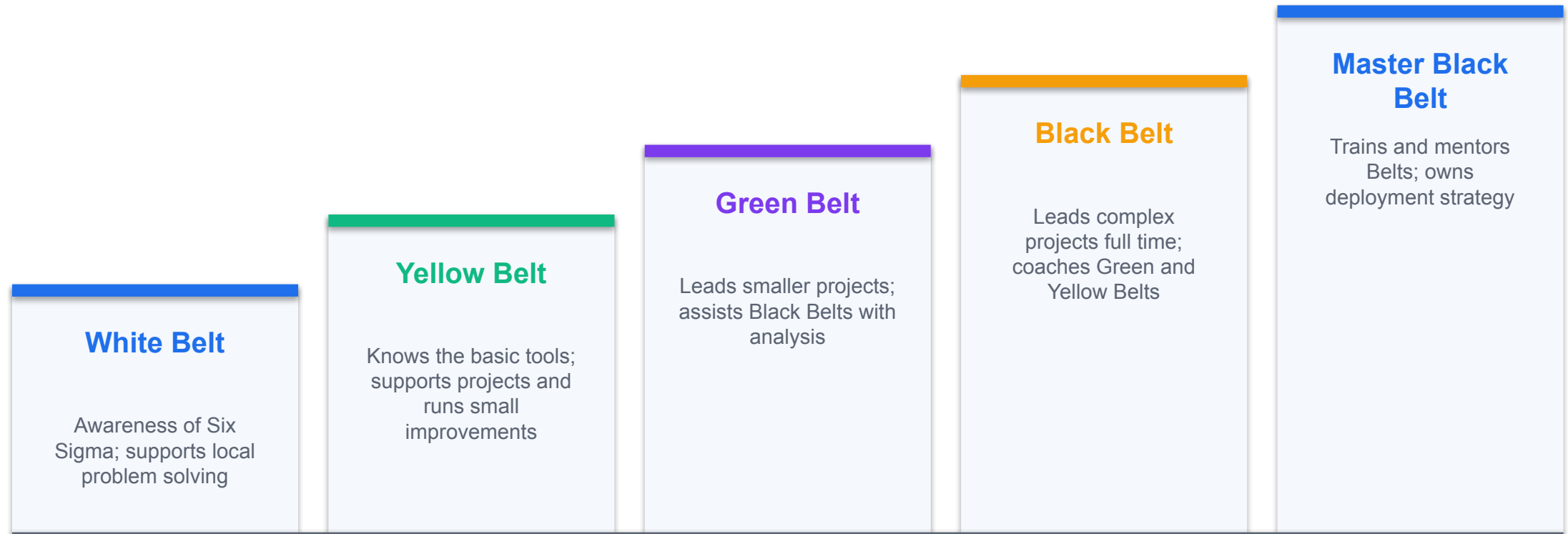


## Lean Six Sigma

Faster AND more consistent — remove the waste, then control the variation that remains.

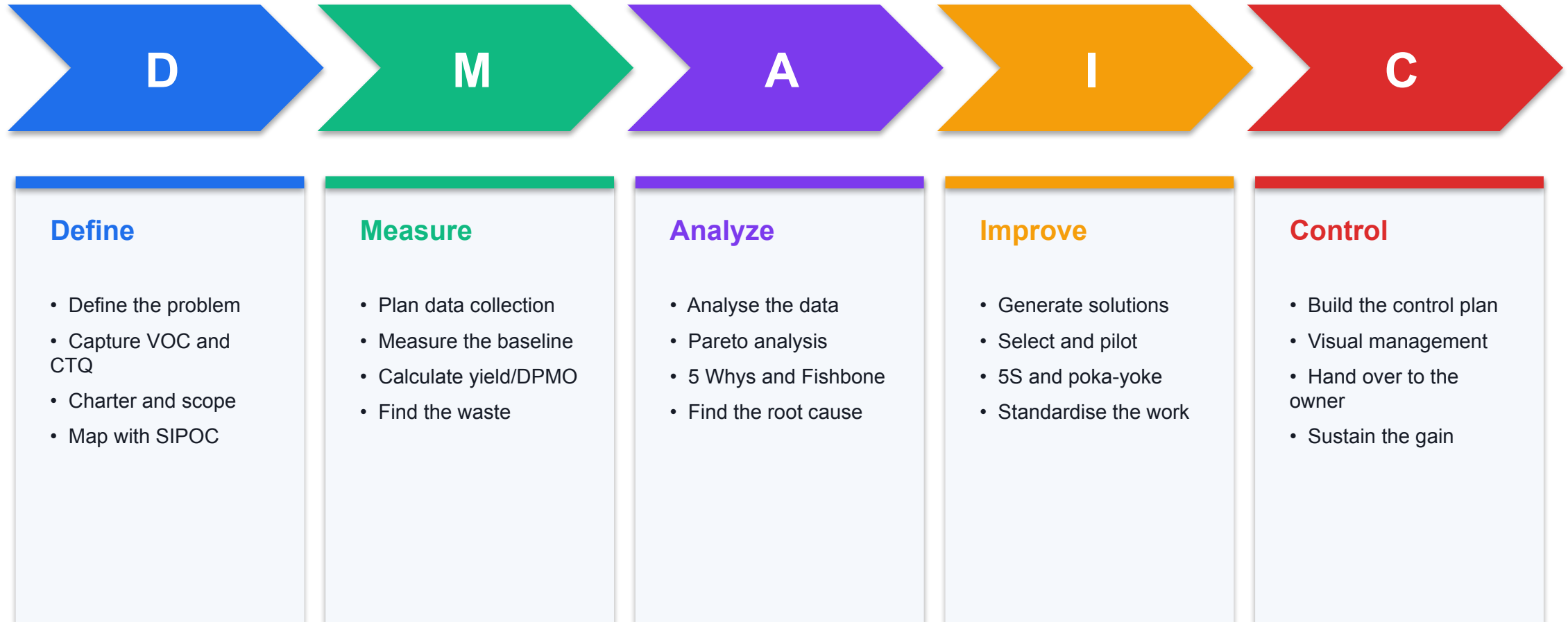


# The Lean Six Sigma Belt Pathway



This course certifies you at White Belt — awareness of Lean Six Sigma and support for local problem solving.

# The DMAIC Roadmap





DMAIC · DEFINE

# Define — Understand the Problem

Voice of the Customer · CTQ · Problem statement · SMART goal · Project charter

# Key Concepts — Define

1

## Voice of the Customer

Ask customers what they need, and record it in their own words.

2

## Critical to Quality

Turn each customer need into something specific you can measure.

3

## Problem statement

What is wrong, where, since when, and how big — never the solution.

4

## SMART goal

Specific, Measurable, Achievable, Relevant and Time-bound.

5

## Project charter

A one-page summary of the problem, goal, scope and team.

6

## Scope

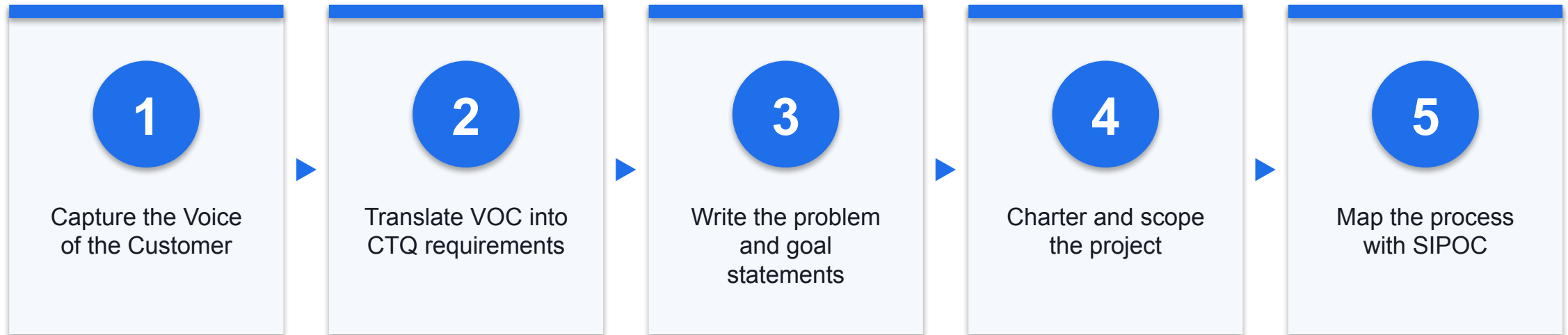
Agreeing what is in and out keeps a small project small.

## DMAIC · D — DEFINE

# Define — What problem are we solving, and for whom?

The Define phase converts a vague complaint into a scoped, measurable problem statement everyone agrees on.

# Steps in the Define Phase



DEFINE · VOC

# Voice of the Customer (VOC)

The needs and expectations of the customer, expressed in the customer's own language.

# Where VOC Comes From

1

## Surveys

Structured and at scale, but limited to the questions you thought to ask.

2

## Interviews

Rich and open-ended; best for understanding the 'why' behind a complaint.

3

## Complaints & reviews

Already coming in unprompted — the cheapest and most honest VOC source.

4

## Direct observation

Go and watch the work happen; customers cannot always articulate the problem.



# VOC → Need → CTQ: The Translation

## Voice of the Customer

- The customer says (VOC)
  - "I don't want to wait too long"
  - "I don't want cold pizza"
  - "I never know how long the wait will be"
  - Vague, emotional, unmeasurable

## Critical to Quality

- The measurable requirement (CTQ)
  - Delivery in 30 minutes or less
  - Pizza at 32°C minimum on arrival
  - Wait time displayed, 5 minutes or less
  - Specific, numeric, testable

# What Makes a Good CTQ?

1

## **Specific**

Names exactly one characteristic of the output.

2

## **Measurable**

Has a number and a unit you can actually record.

3

## **Linked to VOC**

Traces back to something a customer genuinely said.

4

## **Has a target and limit**

States the target value and the acceptable range around it.

# The Project Charter

1

## Business case

Why this project matters and what it is worth.

2

## Problem statement

What is wrong, quantified, over a stated time period.

3

## Goal statement

Metric, baseline, target and date.

4

## Scope

What is in, and explicitly what is out.

5

## Team and roles

Sponsor, project leader, White and Yellow Belt support, and subject matter experts.

6

## Milestones

The dates each DMAIC phase is expected to complete.

# Problem Statements — Good vs Bad

## Avoid this

- Weak problem statement
  - "The morning queue is too slow."
  - No process named
  - No time period
  - No measure or baseline
  - No customer impact stated
  - Hints at a solution ("we need more staff")

## Aim for this

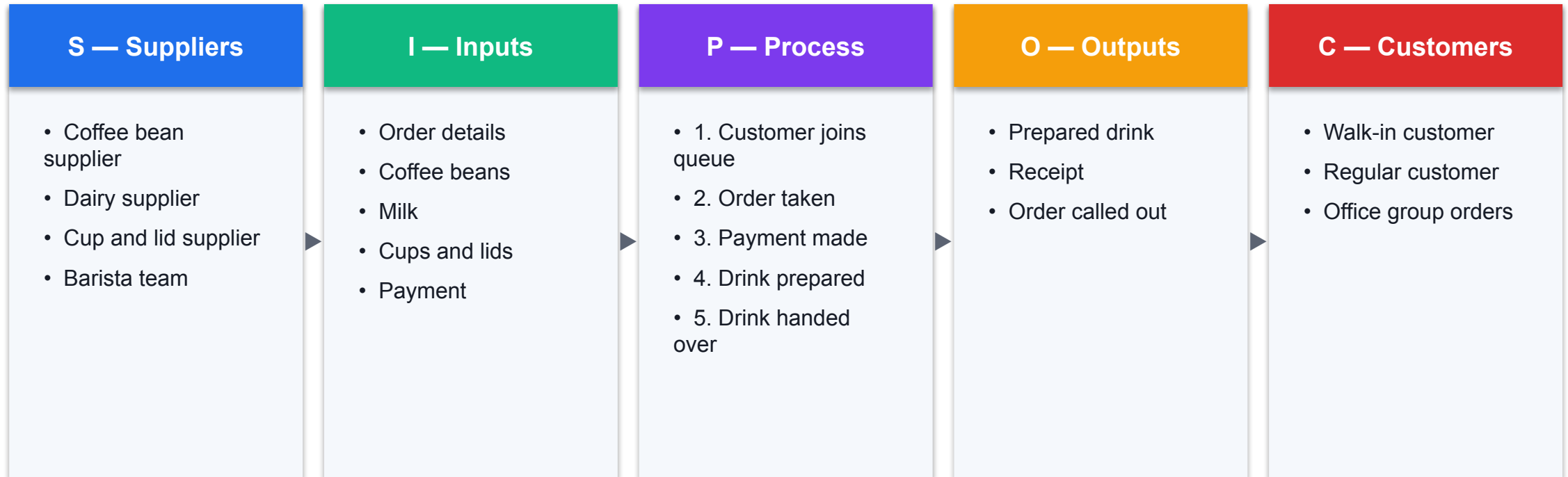
- Strong problem statement
  - "Between 1 and 31 March, customers served between 7:30am and 9:00am waited an average of 11.5 minutes against a 5-minute target, affecting 38% of morning customers and generating 24 online complaints."
  - Process, period, measure, baseline and impact — and no solution.

## DEFINE · SIPOC

# SIPOC — the macro 'as-is' map

One page that shows who supplies the process, what goes in, what happens, what comes out, and who receives it.

# SIPOC — BrewBean Cafe Morning Rush



Read left to right: suppliers provide the inputs, the process transforms them, and the outputs go to the customers.

# Hands-On Activities — Define

1

## Activity 1 — Define — Customer Requirements and the Problem

A VOC-to-CTQ table, a written problem statement, a SMART goal and a si

# Define — Customer Requirements and the Problem Statement

## ACTIVITY 1

Start the BrewBean Cafe improvement the way every real project starts: find out what the customer actually wants, turn that into something measurable, and state the problem clearly enough that everyone agrees on it. The most common reason improvement projects fail is that the team never agreed what the problem was.

### You'll build

**A VOC-to-CTQ table, a written problem statement, a SMART goal and a simple scope table.**

**Tools & techniques:** Voice of the Customer, CTQ, problem statement, SMART goal, project scope



# Your Data Pack — Activity 1

1

## Raw Voice of the Customer feedback

30 rows — voc-raw-customer-feedback.csv

2

## 2 blank templates

Worksheets to fill in — one per step group

3

## A01-Data-Workbook.xlsx

Every dataset and template, one per tab

# Define — Customer Requirements and the Problem Statement

STEPS 1-2 OF 6

1

Read the BrewBean Cafe scenario, then open data/voc-raw-customer-feedback.csv (or the workbook) and read all 30 pieces of customer feedback. Underline every complaint you can find.

2

Using templates/voc-to-ctq-template.csv, write down five Voice of the Customer statements in the customer's OWN words, quoted from the Verbatim Comment column — do not paraphrase them yet.

# Define — Customer Requirements and the Problem Statement

STEPS 3-4 OF 6

3

Turn each VOC statement into a measurable CTQ requirement in the same template — every CTQ must carry a number AND a unit (e.g. 'served within 5 minutes').

4

In templates/problem-statement-and-scope-template.csv, write the problem statement: what is wrong, where, since when, and how big. Do NOT write a solution.

# Define — Customer Requirements and the Problem Statement

STEPS 5-6 OF 6

5

Check your problem statement against the SMART test and rewrite the goal so it is Specific, Measurable, Achievable, Relevant and Time-bound.

6

Fill the in-scope / out-of-scope rows at the bottom of the same template so the team knows where the project stops.

# Define — Customer Requirements and the Problem Statement

## Check your work

Every CTQ has a number and a unit, your problem statement contains no solution, and your goal has a metric, a baseline, a target and a date.

# Recap — Define

- Voice of the Customer — Ask customers what they need, and record it in their own words.
- Critical to Quality — Turn each customer need into something specific you can measure.
- Problem statement — What is wrong, where, since when, and how big — never the solution.
- SMART goal — Specific, Measurable, Achievable, Relevant and Time-bound.
- Project charter — A one-page summary of the problem, goal, scope and team.
- Scope — Agreeing what is in and out keeps a small project small.

DMAIC · MEASURE

# Measure — See What Is Really Happening

Process mapping · SIPOC · Types of data · Data collection · The eight wastes

# Key Concepts — Measure

1

## Process mapping

Draw the steps as they really happen, not as the manual says.

2

## SIPOC

A one-page overview: Suppliers, Inputs, Process, Outputs, Customers.

3

## Types of data

Discrete data you count; continuous data you measure on a scale.

4

## Data collection

Agree what to record, who records it, and when — before you start.

5

## Check sheets

A simple tally form is the easiest reliable way to collect data.

6

## The eight wastes

DOWNTIME — Defects, Overproduction, Waiting, Non-utilised talent, Transport, Inventory, Motion, Extra-processing.

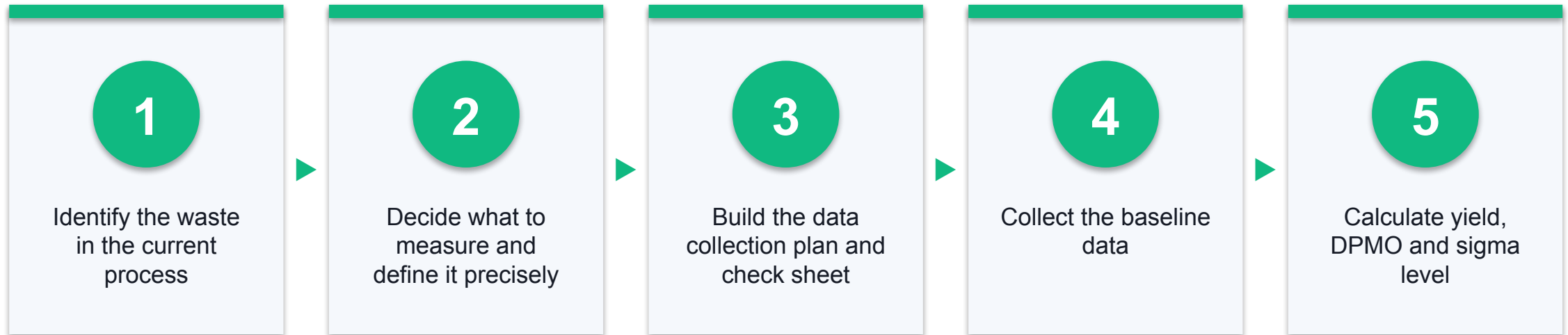


## DMAIC · M — MEASURE

# Measure — How big is the problem, really?

The Measure phase replaces opinion with a trustworthy baseline you can improve against.

# Steps in the Measure Phase

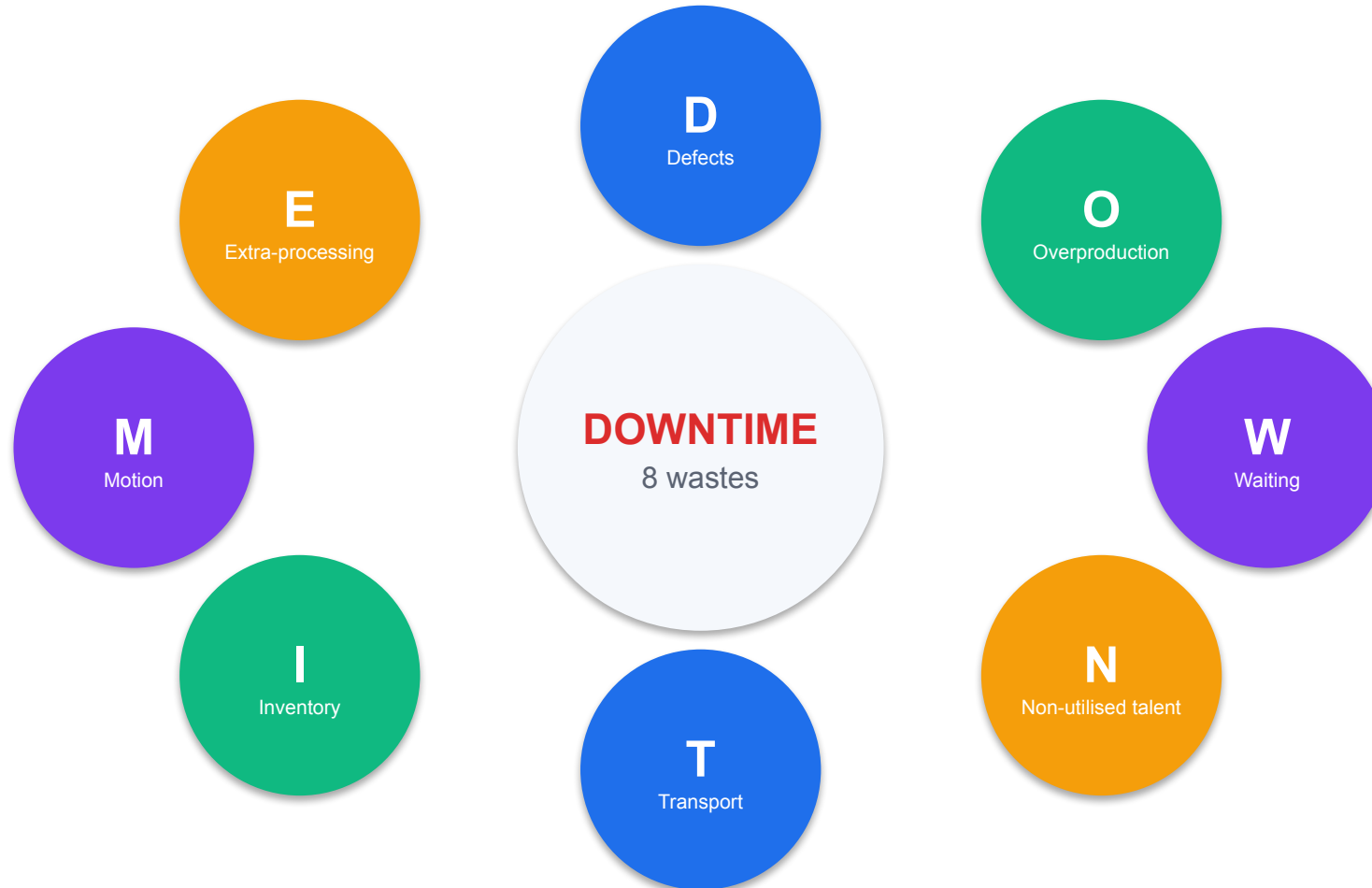


## MEASURE · WASTE (MUDA)

# What is Waste?

Anything beyond the minimum information, equipment, material and effort absolutely required to add value for the customer.

# The Eight Wastes of Lean — DOWNTIME



Waste is any effort the customer would not willingly pay for.

# The Eight Wastes in a Service Process

1

## D — Defects

A drink made wrong and remade; the wrong order handed over.

2

## O — Overproduction

Brewing more coffee at 8am than customers actually order, then pouring it away.

3

## W — Waiting

Customers standing in the queue; a drink waiting on the counter to be called out.

4

## N — Non-utilised talent

An experienced barista tied to the till instead of making drinks at peak.

5

## T — Transport

Carrying milk from the back store to the counter several times each morning.

6

## I — Inventory

A growing line of unmade orders on the rail during the rush.

7

## M — Motion

The barista walking back and forth because cups and lids are stored apart.

8

## E — Extra-processing

Writing the order on a cup and then re-keying it into the till.

# Value-Added vs Non-Value-Added

## Value-added (VA)

The customer would pay for it

- Changes the product or service.
- Done right the first time.
- The customer cares that it happened.
- Example: actually making the coffee.

## Business-value-added (BVA)

Required, but not by the customer

- Needed for legal or regulatory reasons.
- Required to run the business.
- Minimise it — you cannot remove it.
- Example: recording the sale for tax purposes.

## Non-value-added (NVA)

Pure waste — remove it

- Consumes time and resource for nothing.
- The customer would never pay for it.
- Attack this first.
- Example: a customer waiting in the queue.

# Types of Data — and Why It Matters

## Continuous data

Measured on a scale

- Any value within a range.
- Time, cost, temperature, length.
- More information per data point.
- Needs a smaller sample size.

## Discrete / attribute data

Counted in whole units

- Counts and categories only.
- Number of defects, pass/fail.
- Less information per data point.
- Needs a much larger sample.

# The Data Collection Plan

1

## What

Which metric, and what exactly counts as one observation.

2

## Why

Which CTQ or problem statement this metric supports.

3

## How

Automated report preferred; manual capture only when unavoidable.

4

## Who

Someone who knows the process and is available to do it consistently.

5

## When

The frequency and the exact period the data covers.

6

## How many

The sample size — enough to be representative, not so many it is unaffordable.



# Check Sheets — the Simplest Reliable Tool

1

## One row per observation

Never summarise as you collect — record the raw event.

2

## Pre-printed categories

Tick a box rather than writing free text; free text cannot be counted.

3

## Mutually exclusive

Every observation must fall into exactly one category.

4

## Include the context

Date, time, shift and who recorded it — you will need this later.

# Hands-On Activities — Measure

1

## Activity 2 — Measure — Map the Process and Spot the Waste

A SIPOC overview, a step-by-step process map with timings, and a compl

# Measure — Map the Process and Spot the Waste

## ACTIVITY 2

You cannot improve what you cannot see. Walk the BrewBean Cafe morning rush from the customer joining the queue to the drink being handed over, draw it as it really happens, and tally where time and effort are being lost. This is the phase that turns opinions about the problem into facts about the problem.

### You'll build

**A SIPOC overview, a step-by-step process map with timings, and a completed waste tally sheet.**

**Tools & techniques:** SIPOC, process mapping, types of data, check sheets, the eight wastes (DOWNTIME)

# Your Data Pack — Activity 2

1

## Morning rush observation log

60 rows — morning-rush-observation-log.csv

2

## Process step timings

15 rows — process-step-timings.csv

3

## DOWNTIME waste tally (partially completed)

8 rows — downtime-waste-tally.csv

4

## 3 blank templates

Worksheets to fill in — one per step group

5

## A02-Data-Workbook.xlsx

Every dataset and template, one per tab

# Measure — Map the Process and Spot the Waste

## STEPS 1-2 OF 7

1

Agree the start and stop points of the process — start: customer joins the queue; stop: customer receives the drink.

2

Complete templates/sipoc-template.csv: Suppliers, Inputs, Process (5-7 steps only), Outputs, Customers.

# Measure — Map the Process and Spot the Waste

STEPS 3-4 OF 7

3

Open data/process-step-timings.csv. Copy the 15 steps into templates/process-map-template.csv with their observed times.

4

Mark each step VA or NVA in the template — the test is whether the customer would pay extra for THAT step — and write one line saying why.

# Measure — Map the Process and Spot the Waste

STEPS 5-6 OF 7

5

Look at the columns in `data/morning-rush-observation-log.csv` and classify each: which figures are discrete (counted) and which are continuous (measured)?

6

Open `data/downtime-waste-tally.csv` — three rows are completed as worked examples. Using the observation log and the step timings as your evidence, complete the remaining five rows in `templates/downtime-tally-template.csv`.

# Measure — Map the Process and Spot the Waste

## STEP 7 OF 7



Identify which single waste type you tallied most often and at which process step, then write down the baseline figures (average wait, longest wait, % over the 5-minute promise, drinks remade) — Activity 5 measures the improvement against them.



## Measure — Map the Process and Spot the Waste

### Check your work

Your SIPOC has all five columns filled, every process step has a time, and every waste on your tally sheet is tagged to one of the eight DOWNTIME types.

# Recap — Measure

- Process mapping — Draw the steps as they really happen, not as the manual says.
- SIPOC — A one-page overview: Suppliers, Inputs, Process, Outputs, Customers.
- Types of data — Discrete data you count; continuous data you measure on a scale.
- Data collection — Agree what to record, who records it, and when — before you start.
- Check sheets — A simple tally form is the easiest reliable way to collect data.
- The eight wastes — DOWNTIME — Defects, Overproduction, Waiting, Non-utilised talent, Transport, Inventory, Motion, Extra-processing.

DMAIC · ANALYZE

# Analyze — Find the Cause

Root cause · 5 Whys · Fishbone diagram · Pareto chart · Variation

# Key Concepts — Analyze

1

## Symptom vs cause

Treating the symptom makes the problem come back; treating the cause does not.

2

## Root cause

The cause that, once removed, stops the problem recurring.

3

## 5 Whys

Keep asking 'why?' until you reach something you can actually act on.

4

## Fishbone diagram

Sort possible causes into Manpower, Method, Machine, Material and Measurement.

5

## Pareto chart

Roughly 80% of the problem usually comes from 20% of the causes.

6

## Variation

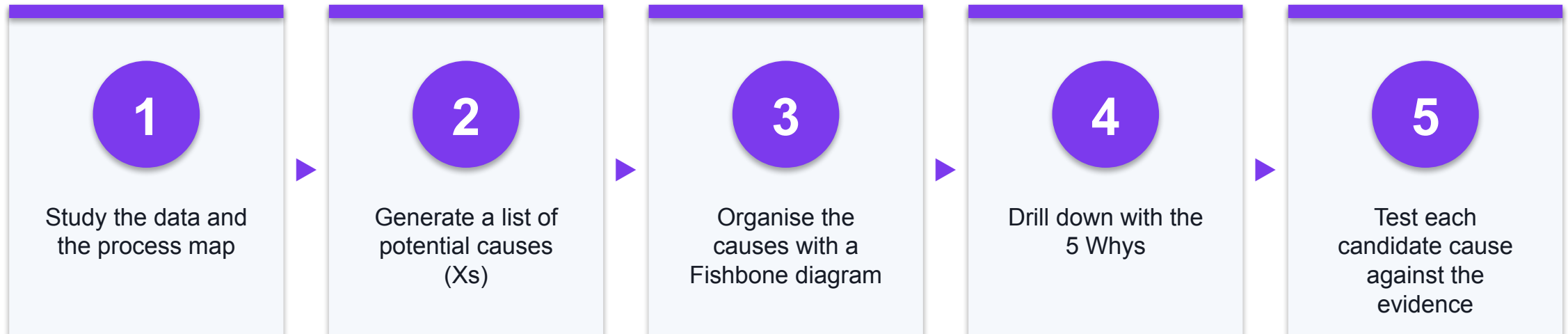
Some variation is normal in the process; some has a specific assignable cause.

## DMAIC · A — ANALYZE

# Analyze — Why is this happening?

The Analyze phase moves the team from a measured symptom to a proven, actionable root cause.

# Steps in the Analyze Phase



# Two Kinds of Variation — and Two Different Responses

## Common cause

Built into the process

- Always present, random, predictable range.
- The process is stable but imperfect.
- Response: change the PROCESS.
- Reacting to individual points makes it worse.

## Special cause

An external, assignable signal

- Unusual, traceable to a specific event.
- Something changed that should not have.
- Response: investigate THAT event.
- Fix the cause, then remove it for good.

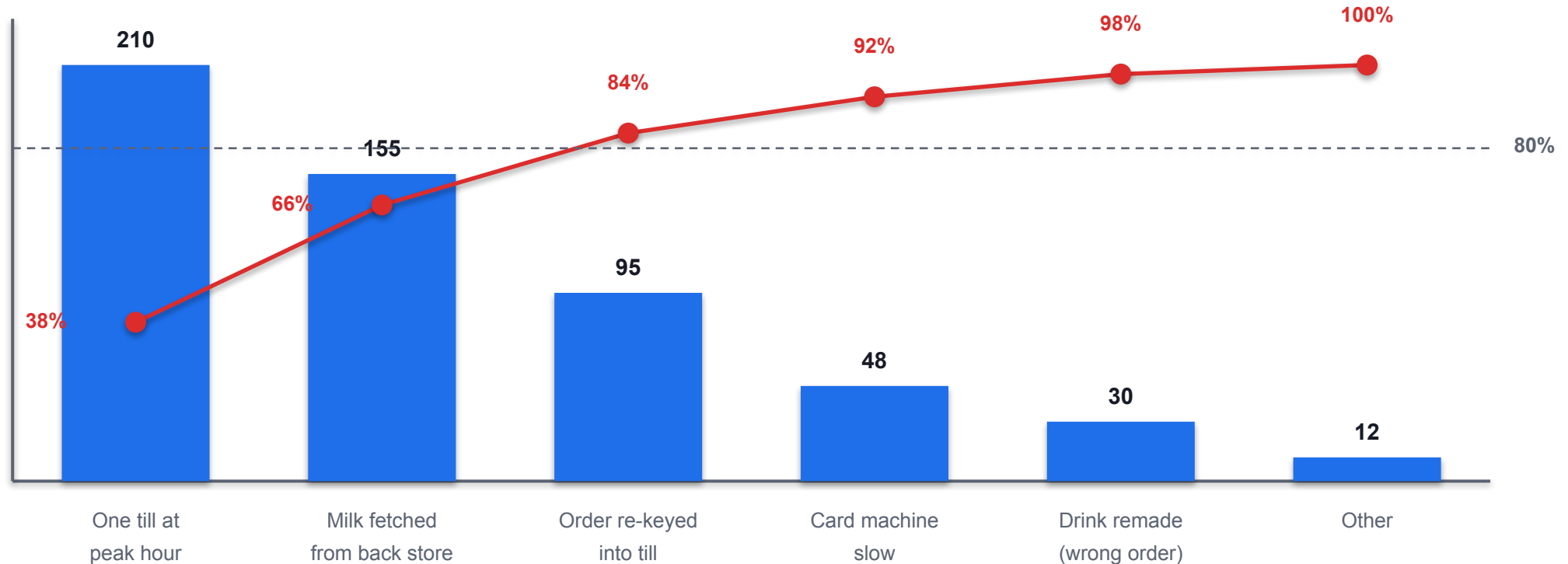
## ANALYZE · PARETO

# The Pareto Principle — the 80/20 rule

Roughly 80% of the effects come from 20% of the causes. Find that 20% and you get most of the improvement for a fraction of the effort.



# Pareto Chart — Causes of Delay at the BrewBean Cafe



Two causes — a single till at peak hour and milk fetched from the back store — account for about 66% of all delay. Start there.

## ANALYZE · ROOT CAUSE

# What is a root cause?

The deepest cause in the chain that you can actually act on — remove it, and the problem does not come back.

# The 5 Whys Technique

1

## Start with the problem

State the effect precisely, using the data you measured.

2

## Ask why — and answer with evidence

Each answer must be something you can point to, not a guess.

3

## Repeat about five times

Five is a guide, not a rule — stop when you reach an actionable cause.

4

## Watch for blame

If an answer names a person rather than a process, ask why again.

5

## Stop at actionable

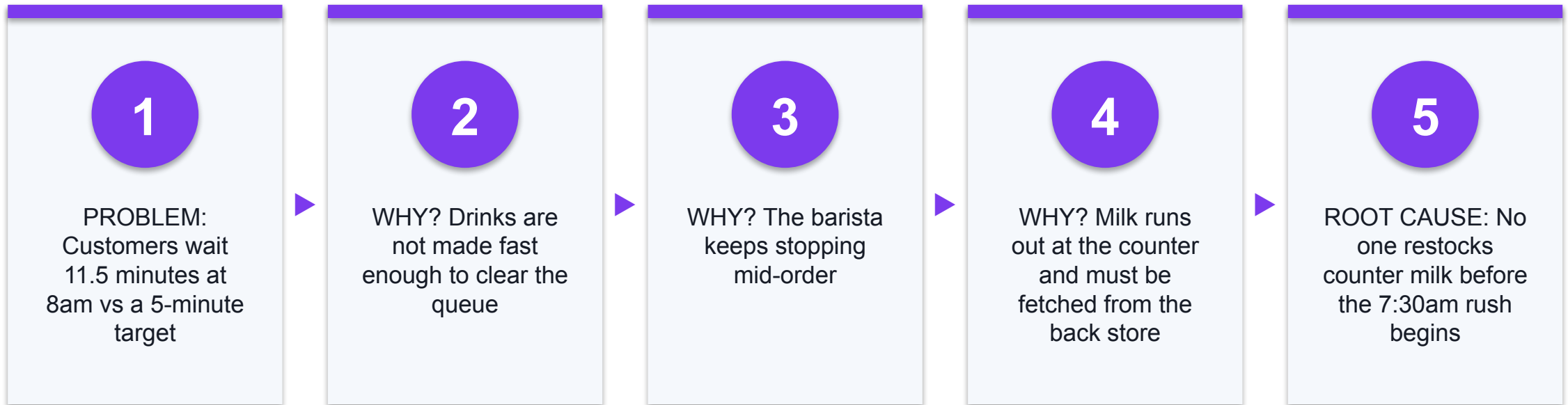
The root cause must be something within the team's power to change.

6

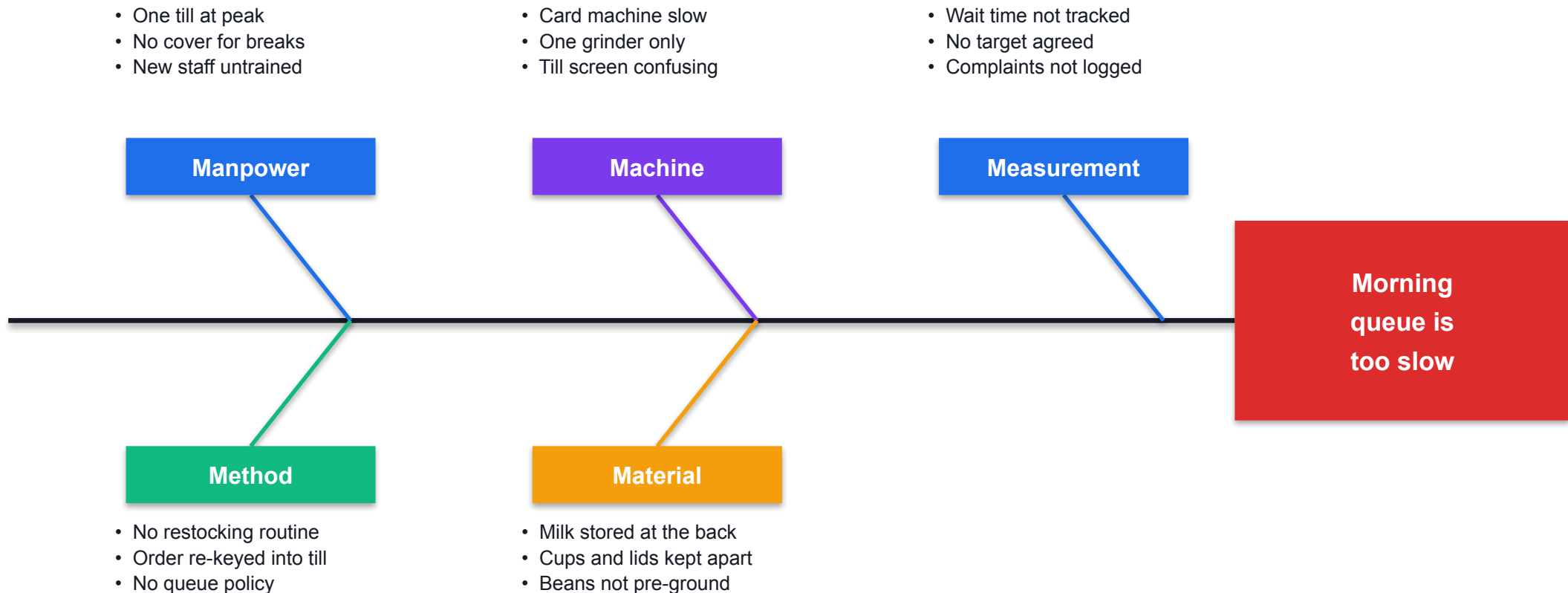
## Verify with data

Test the final cause against the evidence before acting on it.

# 5 Whys — Worked Example: The BrewBean Morning Queue



# Fishbone (Ishikawa) — The Morning Queue Is Too Slow



# Hands-On Activities — Analyze

1

## Activity 3 — Analyze — Find the Root Cause

A completed 5 Whys chain, a Fishbone diagram with causes sorted by cat

# Analyze — Find the Root Cause

## ACTIVITY 3

You now know WHERE the delay happens. This activity asks WHY it happens. Most teams jump straight from a symptom to a solution and end up fixing the wrong thing — the queue is long, so hire more staff — when the real cause was something else entirely. Here you drill past the symptom to a cause you can actually act on.

### You'll build

**A completed 5 Whys chain, a Fishbone diagram with causes sorted by category, and a shortlist of likely root causes.**

**Tools & techniques:** Symptom vs cause, 5 Whys, Fishbone (5M) diagram, Pareto chart, common vs special cause variation

# Your Data Pack — Activity 3

1

## Delay reason Pareto data

6 rows — delay-reason-pareto-data.csv

2

## Candidate causes and their evidence

12 rows — cause-evidence-sheet.csv

3

## 3 blank templates

Worksheets to fill in — one per step group

4

## A03-Data-Workbook.xlsx

Every dataset and template, one per tab



# Analyze — Find the Root Cause

## STEPS 1-2 OF 7

1

Write the symptom from your Activity 2 baseline at the top of templates/5-whys-template.csv (e.g. 'customers wait an average of 8.9 minutes against a 5-minute promise').

2

Ask 'why does that happen?' down the template, each answer becoming the next question. Stop when you reach a PROCESS the team can act on — never at a person.

# Analyze — Find the Root Cause

STEPS 3-4 OF 7

3

Open templates/fishbone-template.csv and write the problem in the head of the fish.

4

Brainstorm possible causes onto the five bones — Manpower, Method, Machine, Material, Measurement — then check them against data/cause-evidence-sheet.csv to see which ones the data actually supports.

# Analyze — Find the Root Cause

## STEPS 5-6 OF 7

5

Open data/delay-reason-pareto-data.csv — the delay reasons from the Activity 2 log, counted and ranked. Read the Cumulative % column and state which few causes account for most of the problem.

6

Using the last column of data/cause-evidence-sheet.csv, decide for each cause whether it is common cause (built into the process) or special cause (a one-off event).

# Analyze — Find the Root Cause

STEP 7 OF 7



In templates/root-cause-shortlist-template.csv, shortlist the two or three causes best supported by evidence — not by opinion.

# Analyze — Find the Root Cause

## Check your work

Your 5 Whys chain ends in something the team can actually act on, every Fishbone cause sits under one of the five categories, and each shortlisted cause is backed by an observation, not an opinion.

# Recap — Analyze

- Symptom vs cause — Treating the symptom makes the problem come back; treating the cause does not.
- Root cause — The cause that, once removed, stops the problem recurring.
- 5 Whys — Keep asking 'why?' until you reach something you can actually act on.
- Fishbone diagram — Sort possible causes into Manpower, Method, Machine, Material and Measurement.
- Pareto chart — Roughly 80% of the problem usually comes from 20% of the causes.
- Variation — Some variation is normal in the process; some has a specific assignable cause.



DMAIC · IMPROVE

# Improve — Fix the Cause

Generating solutions · 5S · Mistake proofing · Standard work · Piloting

# Key Concepts — Improve

1

## Generating solutions

Brainstorm against the proven cause — never against the symptom.

2

## Choosing a solution

Compare ideas on impact, effort and risk before committing.

3

## 5S

Sort, Set in order, Shine, Standardise, Sustain — a tidy workplace for physical or digital work.

4

## Mistake proofing

Poka-Yoke makes the error hard or impossible to make.

5

## Standard work

Write the better method down so everyone does it the same way.

6

## Piloting

Try the change small first, so mistakes stay cheap.

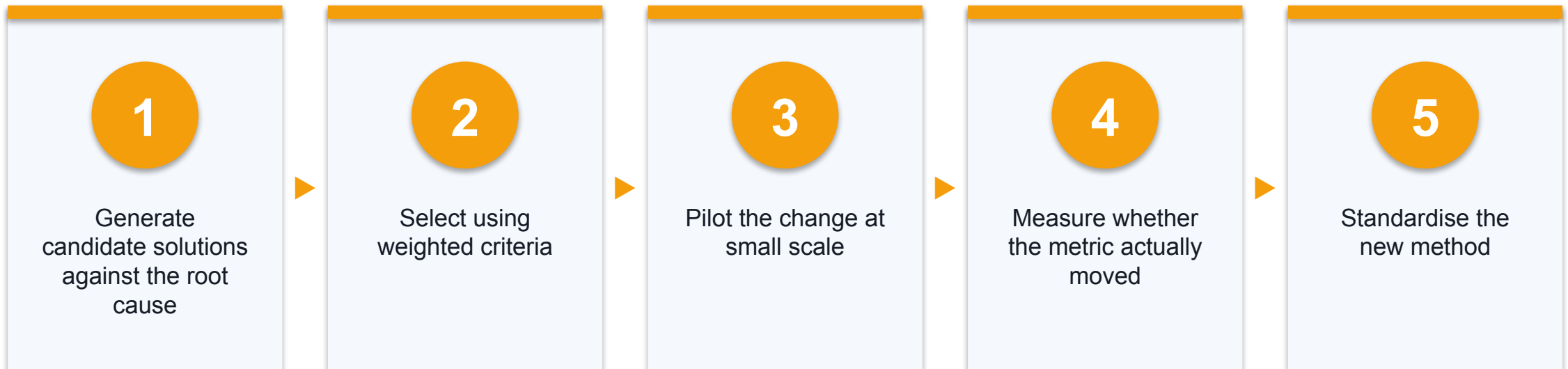


DMAIC · I — IMPROVE

# Improve — What change will actually fix it?

The Improve phase generates, selects and pilots solutions that address the proven root cause.

# Steps in the Improve Phase



# Generating Solutions

1

## Brainstorming

Free generation against the specific root cause — not the general problem.

2

## Brainwriting

Written idea generation; avoids the loudest voice dominating the room.

3

## Benchmarking

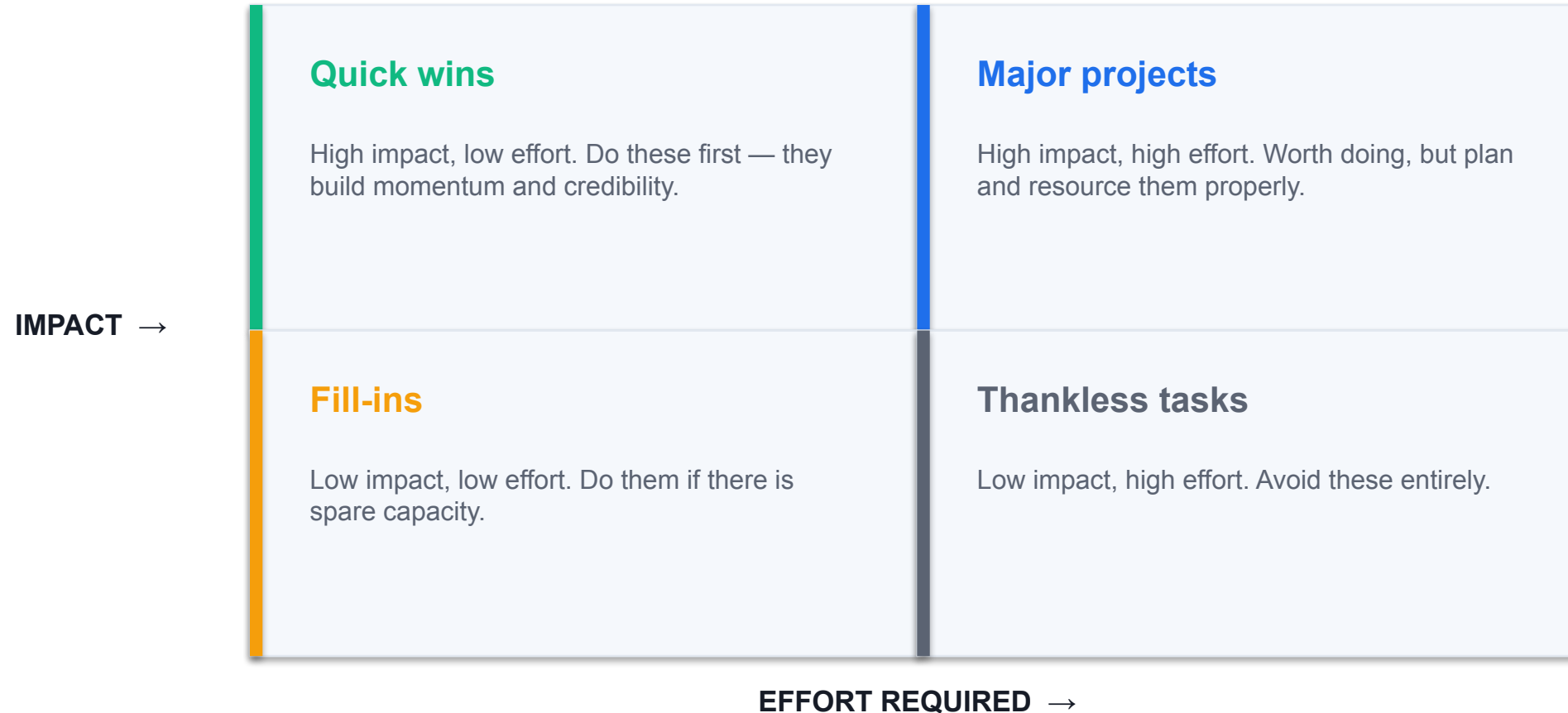
Find who already does this well, internally or externally, and learn from them.

4

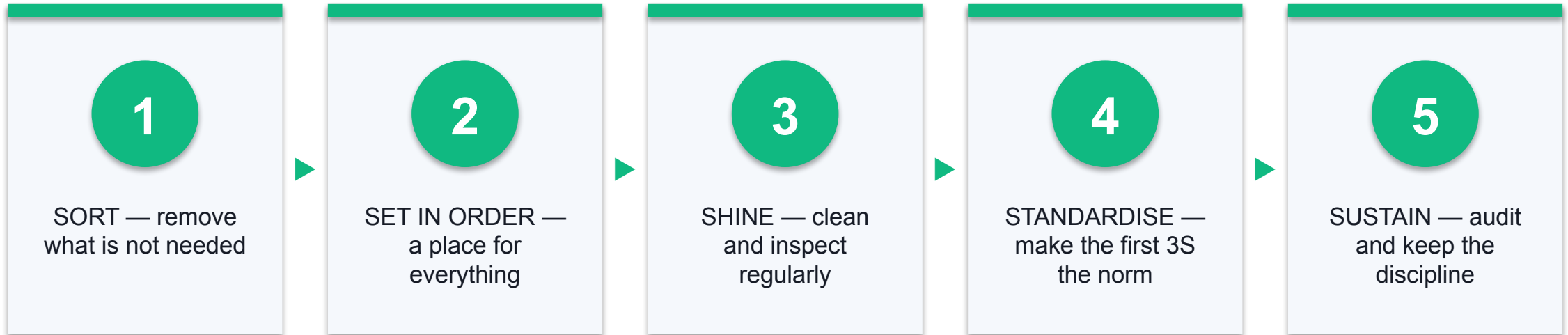
## Anti-brainstorming

Ask how to make the problem worse, then invert every answer.

# Prioritising Countermeasures — Impact vs Effort



# 5S — Organising the Workplace



IMPROVE · POKA-YOKE

# Poka-Yoke — mistake proofing

Design the process so the error is difficult or impossible to make — rather than relying on people to be careful.

# Standard Work

1

## What it is

The documented, current best-known way to perform the task.

2

## Why it matters

You cannot improve a process that everyone performs differently.

3

## What it contains

The sequence, the time each step takes and the quality checks.

4

## Who writes it

The people who do the work — not a manager writing in isolation.

5

## Keep it living

Update it every time a better method is proven.

6

## The Lean insight

Standard work is the baseline for the NEXT improvement, not a straitjacket.

# Piloting Before Full Rollout

1

## Why pilot

Exposes practical issues cheaply, before they affect every customer.

2

## Keep it small

One shift, one counter or one drink type is usually enough.

3

## Measure the same way

Use the identical operational definition as your baseline, or you cannot compare.

4

## Run it long enough

Cover a full business cycle so you see normal variation.

5

## Decide honestly

Adopt, adapt or abandon — a failed pilot that saves a bad rollout is a success.

6

## Then standardise

Only after the pilot proves the gain do you write it into standard work.



## WHAT YOU'LL DO

# Hands-On Activities — Improve

1

### **Activity 4 — Improve — Choose and Pilot a Countermeasure**

A brainstormed countermeasure list, an impact/effort grid, one selecte

# Improve — Choose and Pilot a Countermeasure

## ACTIVITY 4

A cause is only worth finding if something changes because of it. Take the root causes you shortlisted in Activity 3, generate countermeasures against them, and pick one that is small enough to try next week. The discipline here is to solve the CAUSE you proved, not the symptom you started with.

### You'll build

**A brainstormed countermeasure list, an impact/effort grid, one selected countermeasure and a simple pilot plan.**

**Tools & techniques:** Brainstorming, impact/effort screening, 5S, mistake proofing (Poka-Yoke), standard work, piloting

# Your Data Pack — Activity 4

1

## Countermeasure options with impact, effort and cost

12 rows — countermeasure-options.csv

2

## Pilot week results

6 rows — pilot-week-results.csv

3

## 3 blank templates

Worksheets to fill in — one per step group

4

## A04-Data-Workbook.xlsx

Every dataset and template, one per tab

# Improve — Choose and Pilot a Countermeasure

## STEPS 1-2 OF 7

1

Write your top root cause from Activity 3 at the top of templates/countermeasure-brainstorm-template.csv — every idea must address THIS cause.

2

Brainstorm at least six countermeasures without judging any of them yet, then compare your list with data/countermeasure-options.csv, which holds the twelve the team generated.

# Improve — Choose and Pilot a Countermeasure

STEPS 3-4 OF 7

3

Check whether 5S would help: is anything at the counter hard to find, out of place or untidy?

4

Check whether mistake proofing would help: could the wrong order be made impossible rather than merely discouraged?

# Improve — Choose and Pilot a Countermeasure

STEPS 5-6 OF 7

5

Using the Impact and Effort scores in data/countermeasure-options.csv, plot every idea on templates/impact-effort-grid-template.csv and pick ONE from the high-impact, low-effort corner.

6

In templates/standard-work-and-pilot-template.csv, write the improved method as standard work — short numbered steps with an unambiguous done/not-done check.

# Improve — Choose and Pilot a Countermeasure

## STEP 7 OF 7



Complete the pilot plan rows of the same template: what changes, who runs it, what you will measure, the baseline to beat, and how you will know it worked. THEN open data/pilot-week-results.csv and compare what actually happened — pay close attention to Thursday.

# Improve — Choose and Pilot a Countermeasure

## Check your work

Your selected countermeasure clearly addresses the root cause from Activity 3 (not the symptom), and your pilot plan names a measure, an owner and a review date.



# Recap — Improve

- Generating solutions — Brainstorm against the proven cause — never against the symptom.
- Choosing a solution — Compare ideas on impact, effort and risk before committing.
- 5S — Sort, Set in order, Shine, Standardise, Sustain — a tidy workplace for physical or digital work.
- Mistake proofing — Poka-Yoke makes the error hard or impossible to make.
- Standard work — Write the better method down so everyone does it the same way.
- Piloting — Try the change small first, so mistakes stay cheap.

DMAIC · CONTROL

# Control — Hold the Gain

Control plan · Visual management · SOPs · Team huddles · Handover

# Key Concepts — Control

1

## Why Control matters

Without Control, processes quietly drift back to the old way.

2

## Control plan

What we measure, the target, how often, who owns it, and what to do if it slips.

3

## Visual management

Make performance visible so problems are noticed the same day.

4

## Standard operating procedures

Written instructions that lock in the improved method.

5

## Team huddles

Short regular stand-ups catch problems while they are still small.

6

## Handover

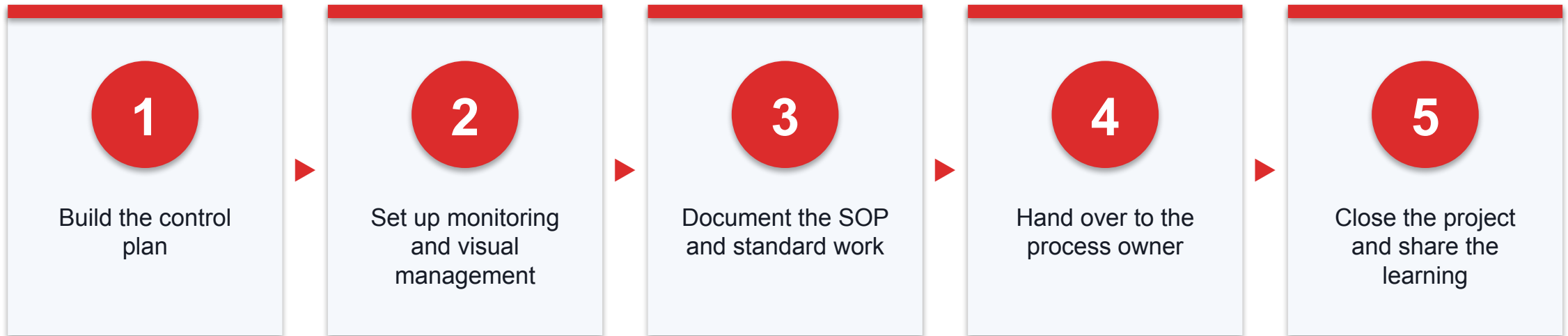
Give the improved process back to the people who run it every day.

## DMAIC · C — CONTROL

# Control — How do we make the gain stick?

Without the Control phase, processes drift back to the old way within months and the whole project is wasted.

# Steps in the Control Phase



# The Control Plan — Six Required Columns

1

## Metric

What is being monitored — the same operational definition as the baseline.

2

## Target

The agreed acceptable value or range.

3

## Method

How the measurement is taken — report, dashboard, sample or audit.

4

## Frequency

How often it is checked: per shift, daily, weekly.

5

## Owner

The named person accountable — a role, never 'the team'.

6

## Reaction plan

Exactly what happens when the metric falls outside target.

# Visual Management

1

## Visual boards

Display the metric, the target and the current status where the team works.

2

## Make it obvious

Anyone should see whether performance is on target within a few seconds.

3

## Team huddles

A short daily stand-up at the board — 15 minutes maximum.

4

## Drives ownership

When performance is visible, the team owns it rather than the manager.

5

## Surfaces issues early

Problems are raised while they are still small and cheap to fix.

6

## Keep it current

An out-of-date board is worse than no board — it teaches people to ignore it.

# Standard Operating Procedures (SOPs)

1

## Written instructions

Describe how to perform the task to achieve the required result.

2

## Include the measures

State the benchmarks and quality checks, not only the steps.

3

## Centrally accessible

Held in one agreed place everyone can reach and search.

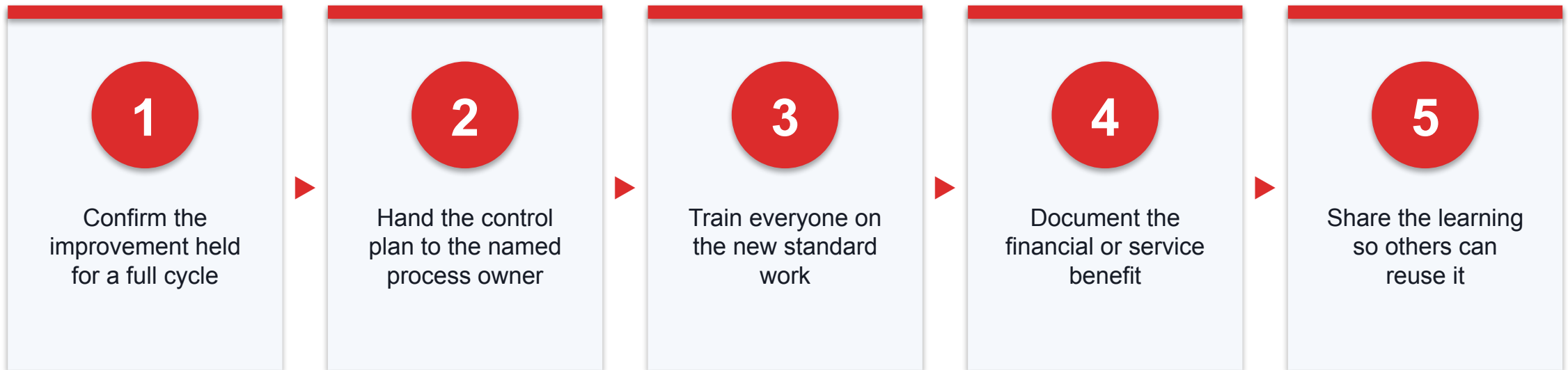
4

## Version controlled

It must be obvious which version is current.



# Project Handover and Closure



## WHAT YOU'LL DO

# Hands-On Activities — Control

1

### **Activity 5 — Control — Hold the Gain and Hand Over**

A completed control plan, a visual management board sketch, and a one-

# Control — Hold the Gain and Hand Over

## ACTIVITY 5

Most improvements work for a fortnight and then quietly disappear. Control is what stops that happening. Close the BrewBean Cafe story by deciding what will be measured from now on, who owns it, what happens when it slips, and how the improved process is handed back to the people who run it every day.

### You'll build

**A completed control plan, a visual management board sketch, and a one-page summary of the whole DMAIC story.**

**Tools & techniques:** Control plan, visual management, SOPs, team huddles, handover, one-page summary

# Your Data Pack — Activity 5

1

## Post-improvement monitoring — four weeks

20 rows — post-improvement-monitoring.csv

2

## Control plan — worked example (one row completed)

4 rows — control-plan-example.csv

3

## 3 blank templates

Worksheets to fill in — one per step group

4

## A05-Data-Workbook.xlsx

Every dataset and template, one per tab

# Control — Hold the Gain and Hand Over

## STEPS 1-2 OF 7

1

Open data/post-improvement-monitoring.csv — four weeks of daily figures after the pilot was made permanent. Choose the ONE measure that tells you the improvement is still working.

2

Look at data/control-plan-example.csv for the worked first row, then complete templates/control-plan-template.csv: measure, target, how often it is checked, where it is recorded and who owns it.

# Control — Hold the Gain and Hand Over

## STEPS 3-4 OF 7

3

Write the reaction plan — the exact steps to take when the measure misses target. Then find week 3 in the monitoring data and state which day your plan would have triggered on.

4

Sketch a simple visual board that would have made the week-3 drift visible to the whole shift on day one.

# Control — Hold the Gain and Hand Over

## STEPS 5-6 OF 7

5

In templates/sop-and-huddle-template.csv, turn your Activity 4 standard work into a short SOP the cafe could actually pin up.

6

Complete the huddle rows of the same template: who attends, what is reviewed, who runs it and how long it lasts.

# Control — Hold the Gain and Hand Over

## STEP 7 OF 7



Using templates/one-page-summary-template.csv, summarise the whole project — problem, cause, countermeasure, result and control — and name who you hand it over to and when.



## Control — Hold the Gain and Hand Over

### Check your work

Your control plan names a measure, a target, a frequency and an owner; your reaction plan says what to DO when it slips; and your one-page summary tells the complete DMAIC story from problem to handover.

# Recap — Control

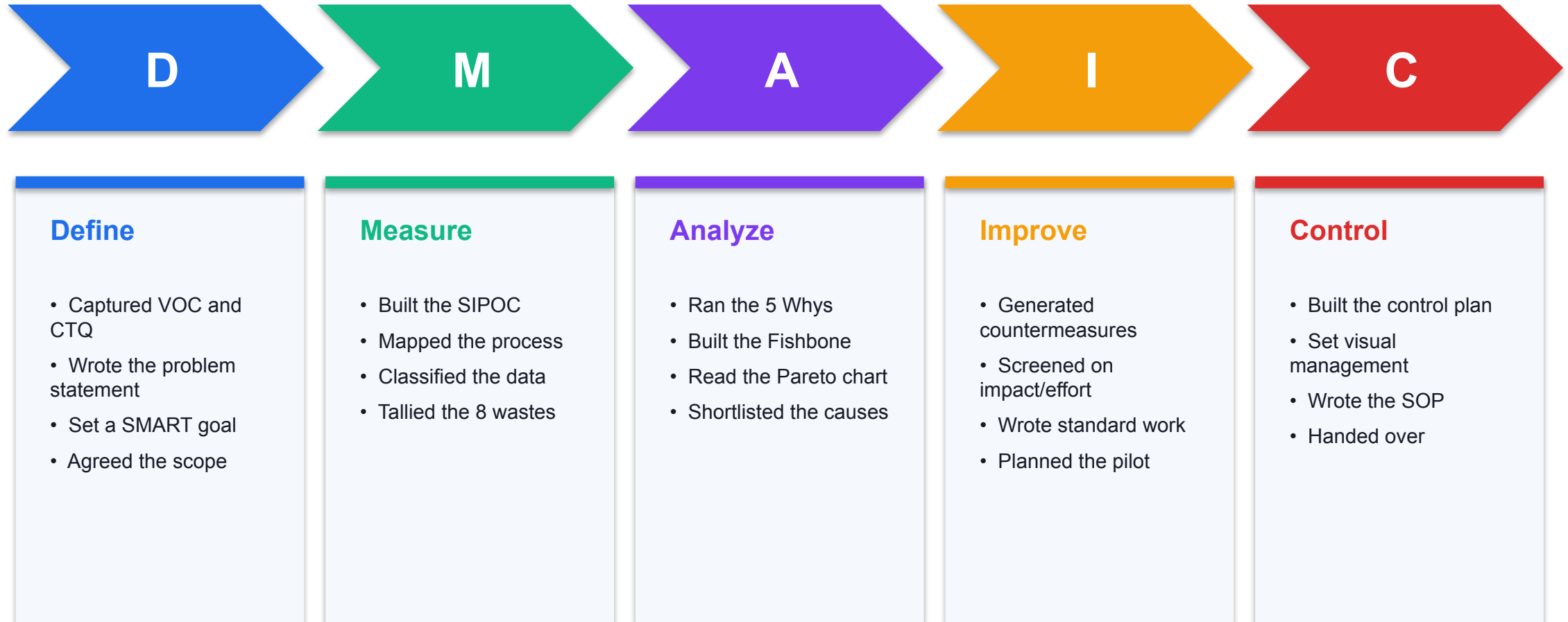
- Why Control matters — Without Control, processes quietly drift back to the old way.
- Control plan — What we measure, the target, how often, who owns it, and what to do if it slips.
- Visual management — Make performance visible so problems are noticed the same day.
- Standard operating procedures — Written instructions that lock in the improved method.
- Team huddles — Short regular stand-ups catch problems while they are still small.
- Handover — Give the improved process back to the people who run it every day.



WRAP-UP

# Course Summary & Next Steps

# What You Achieved — The Full DMAIC Journey



# Your Integrated Improvement Package

1

## VOC and CTQ table

Customer requirements translated into measurable CTQs.

2

## Problem statement and goal

A clear problem statement, a SMART goal and an agreed scope.

3

## SIPOC and process map

The BrewBean Cafe process as it really runs, with times against each step.

4

## Waste tally sheet

Every observed waste tagged to one of the eight DOWNTIME types.

5

## 5 Whys and Fishbone

A cause chain and a categorised cause diagram for your problem.

6

## Countermeasure and pilot plan

One selected countermeasure with standard work and a one-week pilot.

7

## Control plan and SOP

Measure, target, frequency, owner, reaction plan and handover.

8

## One-page summary

The complete DMAIC story from problem to handover.

## BEFORE THE ASSESSMENT

# Final Readiness Checklist

- 1 Can you define quality, Lean, Six Sigma and Lean Six Sigma in your own words?
- 2 Can you name the five DMAIC phases and say what each one delivers?
- 3 Can you describe the belt roles and where the White Belt contributes?
- 4 Can you trace a VOC statement through to a measurable CTQ?
- 5 Can you write a problem statement that contains no solution?
- 6 Can you name the eight wastes and give a workplace example of each?
- 7 Can you explain the difference between a symptom and a root cause?
- 8 Can you run a 5 Whys chain and sort causes on a Fishbone diagram?
- 9 Can you describe what a control plan needs to hold a gain?

## NEXT STEPS

# Continuing Your Lean Six Sigma Journey

1

### Apply it at work

Spot and log the eight wastes in your own area within 30 days.

2

### Yellow Belt

The next step — supports DMAIC projects and the data analysis behind them.

3

### Keep the templates

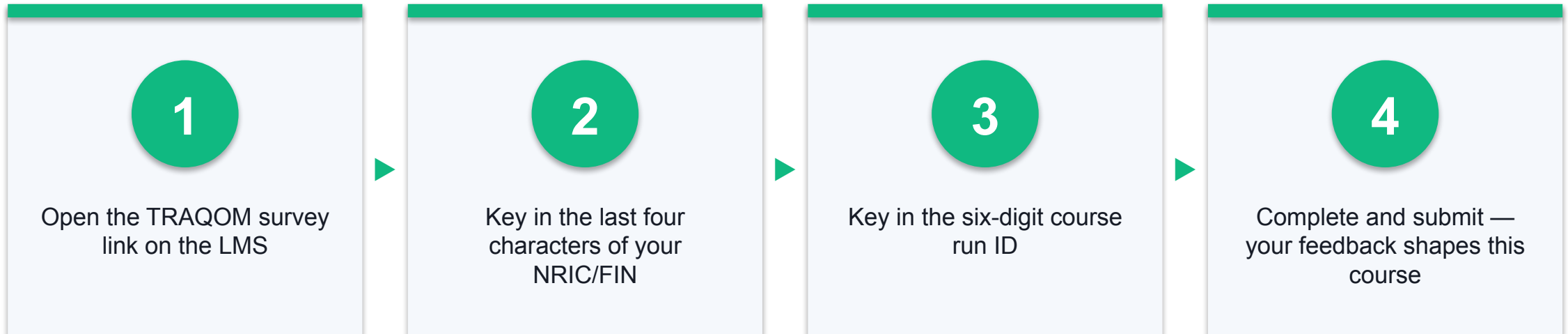
Your lab outputs are reusable templates for real improvement work.

4

### Join the conversation

Raise improvement ideas in your team; small changes spread by example.

# TRAQOM Survey





## Certificate & Support

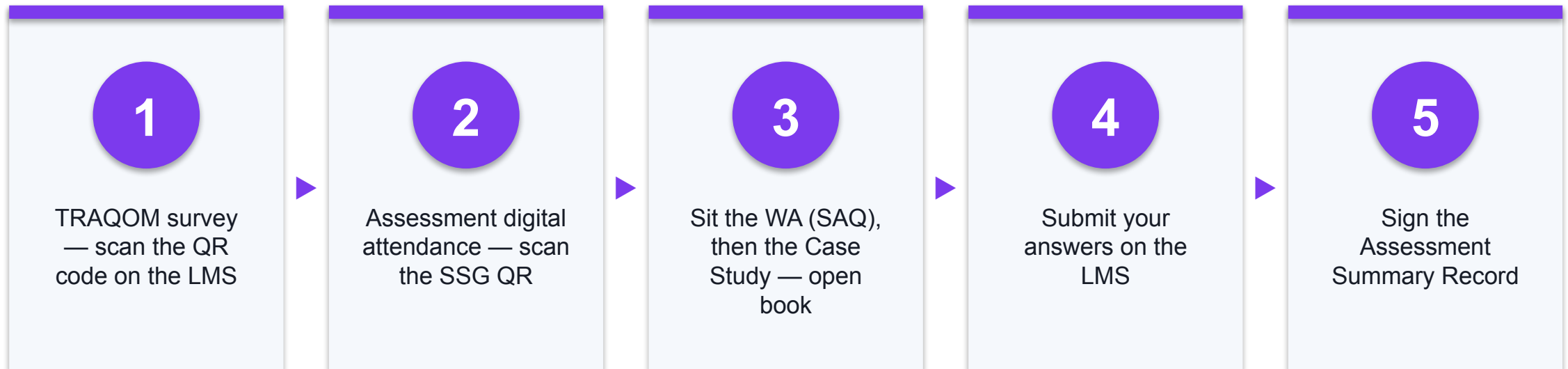
- Two e-certificates are awarded on demonstrating competency and achieving at least 75% attendance.
- A SkillsFuture Statement of Attainment (SOA) is issued for the WSQ assessment.
- Email: [enquiry@tertiaryinfotech.com](mailto:enquiry@tertiaryinfotech.com)
- Tel / WhatsApp: +65 6100 0613

## ASSESSMENT

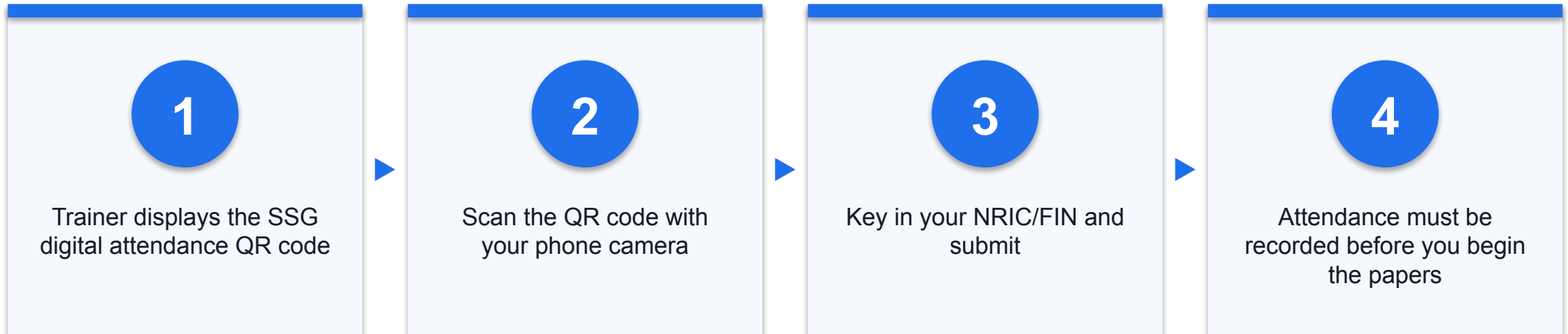
# Final Assessment

Written Assessment (SAQ, 30 minutes) followed by the Case Study (30 minutes). Both are open book.

# Assessment Flow



# Digital Attendance (Assessment)



END OF COURSE

# Thank You!

Go and spot one waste in your own process this month — that is where every improvement starts.